Course Code : CST 364 KQLR/RS – 24 / 1174

Sixth Semester B. Tech. (Electronics and Communication Engineering) Examination

OBJECT ORIENTED DATA STRUCTURE

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory.

(2) All questions carry marks as indicated against them.

(3) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1. (a) Find the output of below Java code :

```
class BoolTest {
    public static void main(String [ ] args) {
        int result = 0 ;
        Boolean b1 = new Boolean("TRUE") ;
        Boolean b2 = new Boolean("true") ;
        Boolean b3 = new Boolean("tRuE") ;
        Boolean b4 = new Boolean("false") ;
        if (b1 = = b2)          result = 1 ;
        if (b1.equals(b2))      result = result + 10 ;
        if (b2 = = b4)          result = result + 100 ;
        if (b2.equals(b4))      result = result + 1000 ;
        if (b2.equals(b3))      result = result + 10000 ;
        System.out.println("result = " + result) ;
    }
}
```

1(CO1)

(b) Show the text output that would be produced by the following main( ) routine :

```
class HelloWorld {
    public static void main(String[ ] args) {
    int n ;     n = 1 ;
    while (n <= 32)
    { n = 2 * n ; System.out.println(n) ; }
    }
}
```

1(CO1)

(c) Write a Java code using **'for'** loop that will print the following series : 3 6 9 12 15 18 21 24 27 30 33 36. 2(CO1)

(d) Explain the Access Specifiers in Java with example. 3(CO1)

(e) Explain the use of user defined package. Create any user defined package and use this package in your java program by importing the package. Can we import same package twice ? Will the JVM load the package twice at runtime ? 3(CO1)

2. (a) For this problem, you should write a very simple but complete class. The class represent a counter that counts 0, 1, 2, 3, 4, ….
The name of the class should be Counter. It has one private instance variable representing the value of the counter. It has two instance methods : increment( ) adds one to the counter value, and getValue( ) returns the current counter value. Write a complete java code. 5(CO2)

(b) Create a class Toy with data members as toyname, category and cost. Include a method set_data( ) to input above information of Toy class and method display data( ) to print the information of entered values. Also include a method print_cost( ) to display the information of toys having cost greater than Rs. 1000. Write a complete Java program to test the class. [Use array of Objects] 5(CO2)

3. (a) Illustrate the uses of super keyword and Final keyword in java with the help of java code. 4(CO1)

(b) Write a Java program that creates a hierarchy for employees of a company. The base class should be **Employee**, with subclasses Manager, Developer and Programmar. Each subclass should have properties such as name, address, salary and job title. Implement methods for calculating bonuses, generating performance reports and managing projects. 6(CO2)

4. (a) Calculate the time and space complexity for the given :

```
int sum(int arr[ ] [n], int n) {
    int i, sum = 0 ;
    for (i = 0 ; i < n ; i + +)
    sum = sum + arr [i] ;
    return sum ;
}
```
2(CO3)

(b) How does one measure the efficiency of algorithms ? Distinguish between best, worst, average case complexities of an algorithm. 3(CO3)

(c) Consider integer array int arr[3][4] declared in 'java' program. If the base address is 1050, find the address of the element arr[2][3] with row major and column major representation of array. 5(CO3)

5. (a) Perform the following operations in a circular queue of length 4 and Give the FRONT, REAR and SIZE of the queue after each operation :

(1) Insert A, B

(2) Insert C

(3) Delete

(4) Insert D

(5) Insert E

(6) Insert F

(7) Delete. 2(CO4)

(b) Convert the following infix expression into postfix format showing stack status after every step in tabular form :

A + (B * C – (D/E ^ F)* G) 3(CO4)

(c) Write a Java method or algorithm to implement a stack using an array with push and pop operations. 5(CO4)

6. (a) Write an algorithm to delete a node after given node in doubly linked list. 5(CO4)

(b) Write a Java code to reverse the elements of a singly linked list.

Input :

Head of following linked list

$1 \to 2 \to 3 \to 4 \to$ NULL

Output :

Linked list should be changed to,

$4 \to 3 \to 2 \to 1 \to$ NULL 5(CO4)